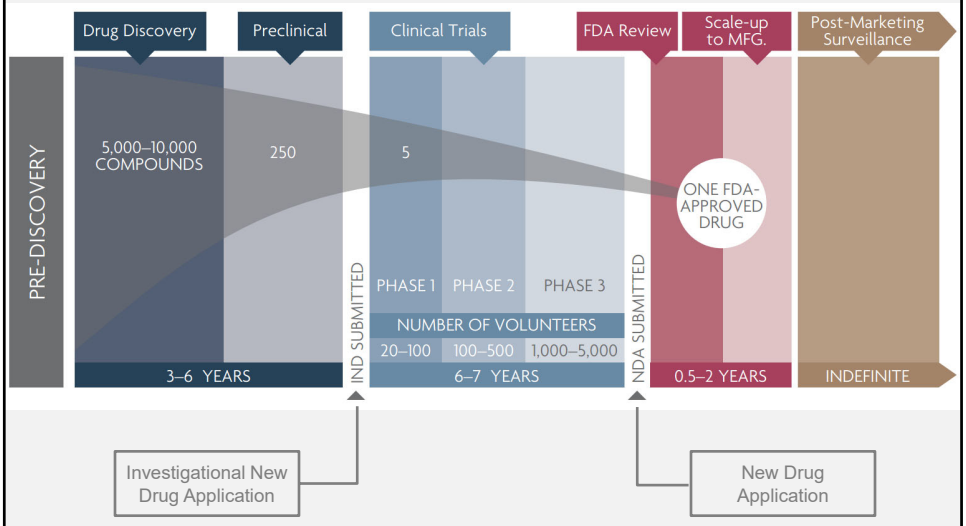
FDA Drug & Device Approval Process

Penn Engineering

EAS5450 Class #22

1

Drug Development Process



2

	Preclinical		Phase I	Phase II	Phase III		FDA	TOTAL
Years	1 - 5 (2.3)		0.5 - 2	1 - 3	1 - 5		1 - 3 (1.8)	5 - 12 (7)
Test population	Laboratory and animal studies	File IND at FDA	20 to 80 healthy volunteers	100 to 300 patient volunteers	1,000 to 3,000 patient volunteers	File NDA at FDA	Review process and approval	
Purpose	Assess safety and biological activity		Determine safety and dosage	Evaluate efficacy, side effects, and dosage	Verify efficacy, monitor adverse reactions from long-term use			
Cost	$0.5 - 5 MM		$5 – 100 MM					$10 - 230 MM
Success rate (Progressing to next step)	18%		75%	57%	80%			0.5 - 7%
	Chemistry, Manufacturing & Controls (CMC) studies		Continued CMC research					

…then Phase IV surveillance to evaluate long-term effects post-marketing.

IND = Investigational New Drug Application
NDA = New Drug Application

Source: Alliance Pharmaceutical Corp. (http://www.allp.com/drug_dev.htm)

3

Drug Development Risk

- **Developing a new drug takes 10-15 years on average.**
- **Only 1 in 10,000 make it.**
- **The average cost to develop one new approved drug may exceed $2.6B.**

Source: Tufts Center for the Study of Drug Development, 2014.

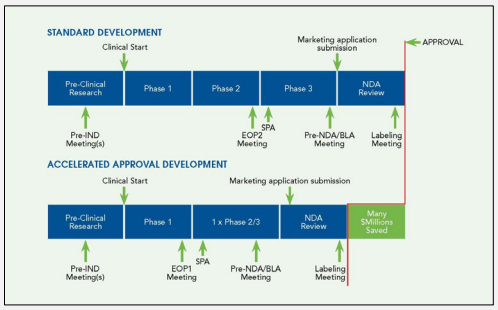
4

Accelerated Approval Mechanisms

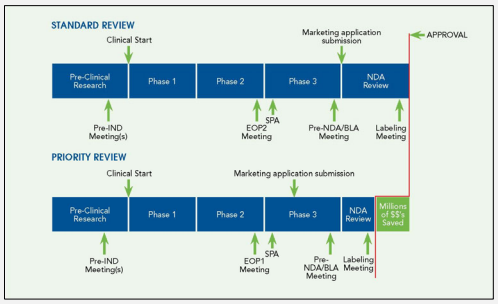
PROGRAM	CRITERIA	INITIATION	BENEFIT
Orphan Drug Act (ODA)	Rare medical conditions. Small populations.	Initiated early in development.	Tax incentives for profitability. Reduced statistical burden. Fast-track FDA approvals.
Expanded Access Program (EAP) *a.k.a. Compassionate Use*	Life threatening with no satisfactory alternative.	Drugs & devices not yet proven safe and effective.	Serves as a clinical trial of sorts.
Accelerated Approval (AA)	Life-threatening condition. Potential for unmet medical need.	Initiated early in development.	Adjusts trial requirements. Requires post-approval clinical studies.
Breakthrough Therapy Designation (BTD)	Life-threatening condition. Clinical evidence shows potential for big improvement over current	Anytime before market approval	Priority status. Submit NDA sections as completed. Rolling review.
Fast Track Designation (FTD)	Life-threatening condition. Potential for unmet medical need. (Same as AA).	Anytime before market approval	Frequent FDA communication. Submit NDA sections as completed. Rolling review.
Priority Review (PR)	Major advance or no adequate existing treatment.	Prior to submitting market application.	Expedites FDA review from 10-12 months to 4-6 months.

5

ACCELERATED APPROVAL



PRIORITY REVIEW



6

FORTUNE

HEALTH • FDA

After 600 emergency use authorizations for COVID, experts worry the FDA has gone too far

BY KAT ESCHNER

June 7, 2021 7:00 PM EDT

SUMMARY:

The Secretary of Health and Human Services (HHS) is issuing this notice pursuant to section 564 of the Federal Food, Drug, and Cosmetic (FD&C) Act. On February 4, 2020, the Secretary determined pursuant to his authority under section 564 of the FD&C Act that there is a public health emergency that has a significant potential to affect national security or the health and security of United States citizens living abroad and that involves a novel (new) coronavirus (nCoV) first detected in Wuhan City, Hubei Province, China in 2019 (2019-nCoV). The virus is now named SARS-CoV-2, which causes the illness COVID-19. On the basis of this determination, he also declared that circumstances exist justifying the authorization of emergency use of drugs and biological products during the COVID-19 pandemic, pursuant to section 564 of the FD&C Act, subject to the terms of any authorization issued under that section.

- Vaccines
  - Pfizer BioNTech
  - Moderna
  - AstraZeneca
  - Janssen (J&J)
- Drugs and Non-Vaccine Biological Products
- Medical devices
  - Blood purification, renal replacement therapy
  - In vitro diagnostics
  - PPE
  - Monitoring products
  - Respiratory assist devices

https://www.federalregister.gov/documents/2020/04/01/2020-06905/emergency-use-authorization-declaration
https://www.fda.gov/emergency-preparedness-and-response/mcm-legal-regulatory-and-policy-framework/emergency-use-authorization#coviddrugs
https://fortune.com/2021/06/07/covid-emergency-use-authorizations-fda-eua-therapies-drugs-medicines-pandemic/

7

Approaches for Venture Capital Funding

- New criteria for companies developing therapeutic drugs and devices

Must have a sustainable financing strategy → AND → Must be significantly better than standard of care → AND → Must Reduce Cost

- Increase emphasis on molecular diagnostics and paired therapies – personalized medicine

DOMAIN ASSOCIATES

8

8

Molecular understanding of disease has shifted industry away from blockbuster model to targeted therapies…

- **1996** - One targeted therapy approved for cancer (Herceptin)
- **2011** - Over 30 targeted therapies approved for cancer (Avastin, Gleevec, etc.)
- "In 10 years it will be a rare drug that does not have a diagnostic test associated with it." - Christopher Seaton, SVP, Bayer

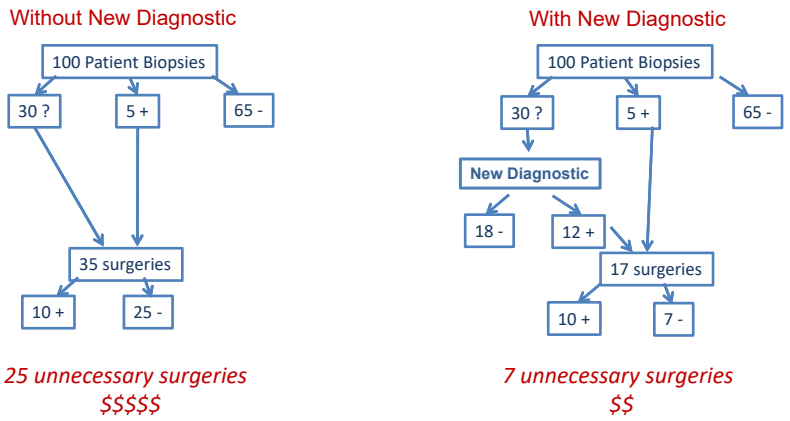
DOMAIN ASSOCIATES

9

9

… products that reduce cost in addition to providing clinical benefit

Suspected Thyroid Cancer Treatment Algorithm



25 unnecessary surgeries
$$$$$

7 unnecessary surgeries
$$

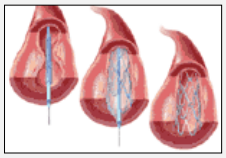
Source: Company publications

DOMAIN ASSOCIATES

10

Devices

- **"Device" refers to**
  - the physical device, and
  - its indication for use
- **Example: Biliary stents**
  - Indication for use: treatment of biliary strictures
  - Use for carotid artery angioplasty represents a new device:
    - Same physical item
    - New indication for use



11

Device Classification

CLASS I
Minimal, if any, patient risk.

CLASS II
Some patient risk.

CLASS III
Supports or sustains human life.

INCREASING RISK

12

Class I Devices

- **General Controls**
  - Labeling regulations
  - Good Manufacturing Practices
  - Registration by manufacturers

Lead shields, radiographic film, bandages, hand-held surgical instruments.

13

Class II Devices

- **General Controls as for Class I**
- **Special Controls:**
  - Performance standards
  - Patient registries
  - Post-market surveillance
  - Testing protocols

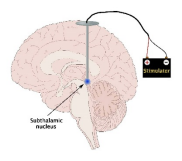
Imaging systems, diagnostic catheters

14

Class III Devices

- **Typical characteristics**
  - Support or sustain human life
  - Risk of illness or injury
- **Scientific review protocol and premarket approval**

Heart pace-makers, heart valves, implanted brain stimulators



15

Medical Device Clearance/Approval

- **510(k) Clearance**
  - Substantially equivalent
  - Faster to market
  - Costs less
  - Weaker I.P. protection (?)
- **Premarket Approval (PMA)**
  - Introduce a device for the 1st time or reintroduce a device
  - Slower to market
  - More expensive
  - Stronger I.P. protection (?)

16

Class II: 510(k) Clearance

- Most common route to market
- Show that new device is “substantially equivalent” to a legally marketed Class I or II device (predicate device)
- Demonstrate that the new device is as safe and effective as the predicate

17

Substantial Equivalence

- Has the same intended use, and
- Has the same technology; or …
- Has different technology, but
  - Does not raise new questions of safety and effectiveness, and
  - Is as safe and effective as the predicate.

18

De Novo Classification

- Novel class I or class II medical devices for which:
  - General controls, or general and special controls, provide reasonable assurance of safety and effectiveness for the intended use; and
  - No predicate device exists.
- May be used as predicates for subsequent devices.

19

Investigational Device Exemptions (IDE)

- Discovery and development of new devices.
- Exempt from controls that apply to devices in commercial distribution.

UNIVERSITY OF PENNSYLVANIA HEALTH SYSTEM

INVESTIGATIONAL RESEARCH BOARD

20

Technology

The FDA Sets Its Sights On Medical Apps

- There are 97,000 health applications available for download
- It takes years "to really master the regulatory system"

In February, **Biosense Technologies Private Ltd.** made a splash when it unveiled uChek, a mobile application and companion kit that lets people use their phone cameras to read subtle color differences on urine test strips. The creators of uChek say their invention can help determine an individual's risk for more than 25 medical conditions, including diabetes and hepatitis. What they didn't anticipate is the need for government approval to sell their product in the U.S.

BusinessWeek: Oct 6, 2013

21

Devices

MOBILE MEDICAL APP (MMA)

PREDICATE

DISTRIBUTED CONFIGURATION

DOCTOR

Used as an accessory to an already regulated medical device

ru

Transforms a mobile device into a regulated medical device

22

FDA approves AliveCor's heart arrhythmia-detecting app algorithm

Above: AliveCor's ECG app.
Image Credit: AliveCor

August 21, 2014 5:30 AM
Mark Sullivan

267 247 18 80 14

While the step counters and run tracker apps of the world get most of the attention, a growing number of apps in the marketplace are serving serious clinical needs.

One is AliveCor's AliveECG app, which will soon run an algorithm that detects atrial fibrillation (AF), a serious type of heart arrhythmia that's often a precursor to stroke. For people with a history or heart trouble, or atrial fibrillation specifically, the app and the algorithm could provide a life-saving early warning system.

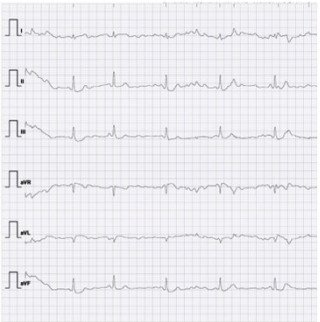
San Francisco-based AliveCor went through a rigorous process with the Food and Drug Administration to prove the accuracy and efficacy of the algorithm. The FDA is particularly interested in reviewing apps that provide any type of actionable, diagnostic information to users, as AliveCor's app clearly does.

The AliveECG app runs on a smartphone. The patient places their hand over an AliveCor nodule device that fits over the back of the phone. The app then measures the electric signals being transmitted by the heart and radiating through the fingers, detected by the nodule.

venturebeat.com/2014/08/21

23

AliveCor KardiaMobile 6L



24

Healthcare IT News

Apple unveils Watch Series 4 with FDA-approved ECG

This is the first FDA clearance for Apple and pushes the Watch further into healthcare than ever before.

By Jonah Comstock | September 12, 2018 | 02:48 PM

- FDA clearance for
  1. Atrial fibrillation-detecting algorithm
  2. ECG built into Apple Watch Series 4
- De Novo clearance ("first of its kind")
- "Placing your finger on the digital crown creates a closed circuit with electrodes on the back"
- Irregular rhythm not intended to be used by those previously diagnosed with atrial fibrillation (~best for those who are "well")

25

FitBit Sense 2

26

Electrophysiology Lab

Penn Presbyterian Hospital

27

FDA Advice…

- Early FDA consultation aids in the development and approval process.
- Consider the FDA a development partner.
- FDA approval does not guarantee insurance reimbursement
  - Need clinical efficacy
  - Economic pricing model

28

sirtris

Sirtris Pharmaceuticals

Operations Strategies

29

Exhibit 2 Sirtris Scientific Advisory Board

BioPharma Expertise	Sirtuin Expertise
Tom Salzmann, Co-Chair SAB Former EVP Merck	Biochemistry John Denu, Wisconsin Anthony Sauve, Cornell Vern Schramm, AECOM David Sinclair, Co-Chair SAB, HMS Eric Verdin, UCSF
Julian Adams President and CSO, Infinity	
Peter Hutt Former FDA General Counsel	Mouse Genetics, Phenotypes Fred Alt, HMS, NAS, HHMI Johan Auwerx, ICB
Bob Langer MIT, NAS, NAE, IOM, Co-Founder Momenta, AIR	Structural Biology Cynthia Wolberger, JHU, HHMI
Tom Maniatis Harvard, NAS, Co-Founder GI, Acceleron	Links to Disease Hon Kahn, Joslin, NAS Jeff Milbrandt, Washington University Jerrold Olefsky, UCSD Pere Pulgserver, HMS Eric Ravussin, Pennington Institute Li-Huei Tsai, MIT, HHMI
Phil Sharp Co-Founder Biogen and Alnylam, NAS, Nobel Prize	
Ted Sybertz SVP Genzyme, Schering (Zetia)	
Eric Gordon Former head of Medicinal Chemistry at Bristol-Myers, Founder of Vicuron Pharmaceuticals	
Chris Walsh Harvard, NAS, IOM, Genzyme, Vicuron	

Source: Company.

HHMI = Howard Hughes Medical Institute
ICB = Mouse Clinical Institute in Strasburg, France
NAS = National Academy of Sciences
NAE = National Academy of Engineers
AECOM = Albert Einstein College of Medicine

30

Exhibit 3 Sirtris Board of Directors (as of February 2007)

Name	Background
Richard Aldrich (F)	Founder, RA Capital
John Clarke	Managing General Partner, Cardinal Partners
Alan Crane	Momenta
John Freund	Managing Director, Skyline Ventures
Stephen Hoffman, Ph.D., M.D.	Managing Director, TVM Capital
Wilf Jaeger	Three Arch Partners
Stephen Kraus	Bessemer Venture Partners
Richard Pops (F)	Chairman, Alkermes
Paul Schimmel, Ph.D. (F)	Professor, The Scripps Research Institute
David Sinclair, Ph.D. (F)	Associate Professor, Harvard Medical School
Christoph Westphal, M.D., Ph.D. (F)	CEO and Vice Chair, Sirtris Pharmaceuticals, Inc.

Source: Company.

F = Founder

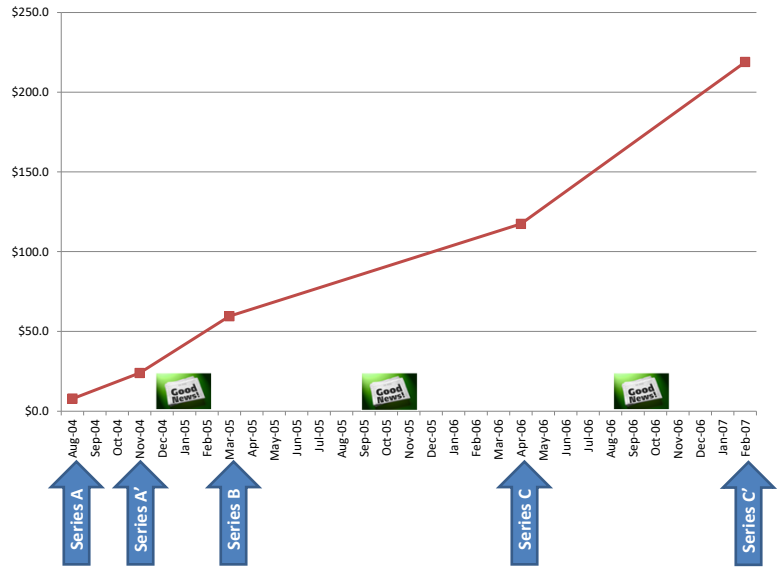
31

Sirtris Investment Cap Table

			RAISED	POST-MONEY VALUATION	% SOLD	FOUNDERS' OWNERSHIP	FOUNDERS' "VALUE"
Aug-04	Series A	4 VC's	$5,000,000	$7,800,000	64.1%	35.9%	$2,800,000
Nov-04	Series A'	6 VC's	$13,000,000	$23,900,000	54.4%	16.4%	$3,900,000
Mar-05	Series B	9 VC's	$27,000,000	$59,500,000	45.4%	8.9%	$5,300,000
Apr-06	Series C	15 VC's	$22,000,000	$117,400,000	18.7%	7.3%	$8,500,000
Feb-07	Series C'	VC's + Ind	$35,900,000	$219,900,000	16.3%	6.1%	$13,400,000

32

Sirtris Post-Money Valuations



33

Strategic Options Facing Sirtris

1. Neutraceuticals
2. Partnering With Pharma
3. In-Licensing

34

Potential Sirtris Pharma Deal Terms

- Five year term.
- 19.9% OR 51% equity stake at a 50% premium to most recent share price.[a]
- $75 million upfront for an exclusive option to join Sirtris in developing and marketing compounds from its SIRT1 Activator Program.
- 5 years of guaranteed R&D support totaling $100 million. Payments to step up over time. $10 million in year one.
- A combination of royalties, possibly manufacturing profits, and co-promotion fees that equate to approximately a 50/50 split of profits in the U.S. This is a significant point for Sirtris since the SIRT 1 activator program is a core program, and the one that represents 90% plus of the firm's value.
- The pharmaceutical company will lead marketing and country specific development ex-U.S., Sirtris to receive substantial, double digit royalties on ex-U.S. sales.
- Roughly $200 million in milestones concurrent with risk reducing progress. Roughly 15% upon successful completion of safety/PK of a SIRT 1 activator NCE in humans; Roughly 25% based on observation of glucose effects in phase 1b of NCE in man; Roughly 30% upon successful completion of a phase 2a efficacy study for an NCE in man; and roughly 30% upon completion of phase 2b studies.

a: Two different equity stakes were under discussion—19.9% or 51% of Sirtris's equity. All other terms would remain the same in both scenarios.

35

TechConnect

About Community Events News Publications Membership

Sirtris Pharmaceuticals In-Licenses Key Genetic Driver of Cellular Stress Resistance and Longevity in Mammals from Robarts Research Institute

March 12, 2008 08:00 AM US Eastern Timezone

Sirtris Pharmaceuticals, Inc., a biopharmaceutical company focused on discovering and developing small molecule drugs to treat diseases of aging such as Type 2 Diabetes, announced today that it obtained exclusive, worldwide rights from Robarts Research Institute (London, Ontario, Canada) for NamPRT (also known as Visfatin), an important genetic driver of cellular stress resistance and longevity in mammals.

"In-licensing the exclusive rights to these patents is part of our overall strategy to build the broadest platform of technology and intellectual property for the development of sirtuin therapeutics for the treatment of diseases of aging," said Karl Normington, PhD, Senior Director of Intellectual Property and Licensing, Sirtris Pharmaceuticals. "This intellectual property expands our existing portfolio around the NAD pathway, including nicotinamide and nicotinamide riboside, as well as other genes known to regulate intracellular NAD levels."

"We believe our internal efforts, combined with access to technology from premier research institutions such as Harvard University, Massachusetts Institute of Technology, and Robarts Research Institute, will continue to keep Sirtris at the forefront of the sirtuin therapeutics field," said Christoph Westphal, MD, PhD, Chief Executive Officer and Vice Chair of Sirtris Pharmaceuticals.

36

Sirtris Investment Cap Table

			RAISED	POST-MONEY VALUATION	% SOLD	FOUNDERS' OWNERSHIP	FOUNDERS' "VALUE"
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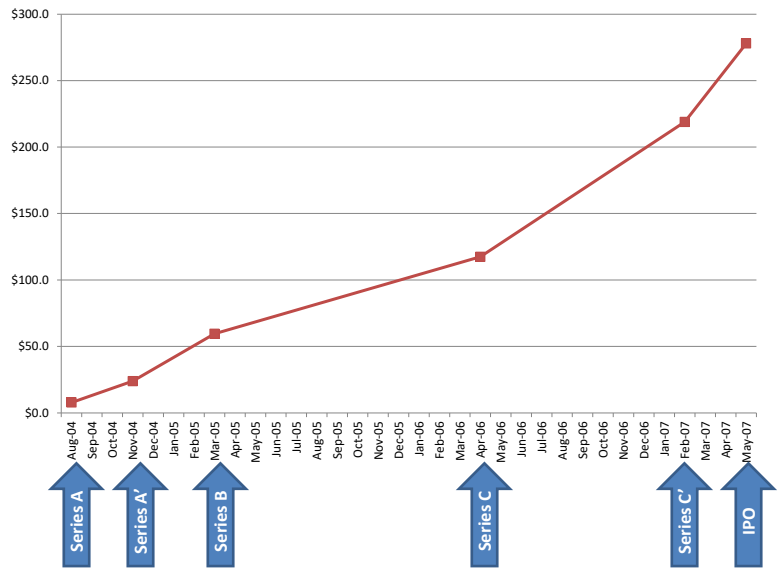
Feb-07	Series C'	VC's + Ind	$35,900,000	$219,900,000	16.3%	6.1%	$13,400,000
May-07	IPO		$55,800,000	$278,000,000	21.6%	4.8%	$13,300,000

37

Sirtris Post-Money Valuations



38

GSK Acquisition Details

- April 22, 2008
- $22.50/Shr = 80% premium over $12.50/Shr market price
- $715.6 million valuation
- 25% 4-year holdback provision

	Original Common Shares	Vested Options	Accelerated Options	Accelerated Restricted Stock	TOTAL SHARES	TOTAL VALUE ($22.50/Sh)
Christoph Westphal	355,114	173,256	409,217	207,136	**1,144,723 (3.6%)**	**$25.76MM**
David Sinclair	121,508	121,221	97,559	31,326	**371,614 (1.2%)**	**$8.36MM**

Source: SEC Schedule 14D-9 for Sirtris Pharmaceuticals

39

GSK Acquisition Holdback Provision

Subject to specified exceptions, the cash payments described in this paragraph are subject to holdback arrangements designed to encourage retention of Company personnel, to the extent such payments are in respect of Options or Restricted Stock unvested at the Purchase Time. Under the holdback arrangements, 25 percent of the after-tax amount of such payment will be placed into a custodial brokerage account invested in GSK common stock and will be paid out over four years from the Purchase Time. If the recipient voluntarily resigns, any remaining payments from the custodial account will be forfeited.

Sirtris SEC Filing: Schedule 14D-9, filed May 2, 2008.

40

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sirtris
A GSK COMPANY

ABOUT US | DISCOVERY | PIPELINE | PUBLICATIONS & MEETINGS | CULTURE

Translating Sirtuin Science into Revolutionary Medicines for a Healthier World

Click here to learn how activating SIRT1 offers broad potential for **diseases of aging**

Welcome to Sirtris

Sirtris is focused on discovering and developing proprietary, small molecule drugs with the potential to treat diseases associated with aging, such as chronic disorders involving metabolic and inflammatory dysfunction and neurodegeneration. Our research focus is on modulating the sirtuins, a class of enzymes demonstrated to be involved in multiple diseases associated with advancing age.

gsk GlaxoSmithKline
Sirtris, a GSK company
www.gsk.com

41

THE WALL STREET JOURNAL.
WSJ.com

HEALTH INDUSTRY | MAY 5, 2010

Glaxo's 'Red Wine' Drug Trial Halted Over Safety Concerns

By JEANNE WHALEN And PETER LOFTUS

In a possible setback for scientists attempting to make drugs out of a substance found in red wine, GlaxoSmithKline PLC said a clinical trial of one drug in cancer patients has been halted due to safety concerns.

Glaxo acquired the drug in 2008 when it paid $720 million for Sirtris Pharmaceuticals, a biotech company in Cambridge, Mass. The drug, known as SRT501, contains a reformulated version of resveratrol, a substance found in low quantities in red wine.

Glaxo and Sirtris have been testing the drug and others in several diseases, including diabetes and cancer, in the belief that they may provide health benefits by activating enzymes in the body called sirtuins.

The suspension of the trial comes amid recent debate in scientific journals about how well the experimental drugs work.

Glaxo said Tuesday it has halted a small trial in multiple myeloma, a cancer of the plasma cells in bone marrow, because some patients developed a complication called cast nephropathy, a condition that can cause kidney failure.

George Vlasuk, chief executive of Sirtris, said in a phone interview that the complication is common in multiple myeloma patients and "may or may not be related to" the drug. Glaxo halted the trial "out of an abundance of caution" while it studies the data further, he said.

42

Published on FierceBiotech (http://www.fiercebiotech.com)

New Study Validates Longevity Pathway

March 7, 2013 | By Jennifer Levin

New Study Validates Longevity Pathway
Findings identify universal mechanism for activating anti-aging pathway

FINDINGS
A new study demonstrates what researchers consider conclusive evidence that the red wine compound resveratrol directly activates a protein that promotes health and longevity in animal models. What's more, the researchers have uncovered the molecular mechanism for this interaction, and show that a class of more potent drugs currently in clinical trials act in a similar fashion.

RELEVANCE
The findings not only put to rest questions about how resveratrol and related compounds work, but also will help scientists better engineer molecules that more precisely and effectively treat disease associated with aging.

PRINCIPAL INVESTIGATOR
David Sinclair, Harvard Medical School professor of genetics
http://www.hms.harvard.edu/dms/bbs/fac/sinclair.php

JOURNAL
Science

The Boston Globe

By Carolyn Y. Johnson | GLOBE STAFF MARCH 07, 2013

Outside researchers were less enthusiastic. They said the new paper provided a plausible and more nuanced explanation for how resveratrol works, but will do little to clarify the bigger questions in the murky field. The role of a particular sirtuin, called SIRT1, in aging, "is still as clear as mud," said Brian Kennedy, president of the Buck Institute for Research on Aging, a nonprofit research center in Novato, Calif.

43

[Updated 2:25 pm ET, 3/13/13] GlaxoSmithKline is closing down Cambridge, MA-based Sirtris Pharmaceuticals, almost five years after it paid $720 million to acquire the hot biotech with a plan to fight diseases of aging.

London-based Glaxo (NYSE: GSK) is shutting the Cambridge, MA Sirtris facility, and plans to offer transfers to the Philadelphia area for some of the 60 remaining Sirtris employees, according to news first reported today by FierceBiotech. Glaxo plans to review staff in coming weeks and decide who will get the opportunity to transfer, GlaxoSmithKline spokeswoman Melinda Stubbee says. Aside from the Sirtris operation, GSK still has staff in Waltham, MA and a few others in Cambridge, Stubbee tells Xconomy.

www.xconomy.com

44

FierceBiotech
THE BIOTECH INDUSTRY'S DAILY MONITOR

Christoph Westphal launches another biotech, with the help of some wealthy friends

September 8, 2014 | By John Carroll

Christoph Westphal

Proving once again that Christoph Westphal has some of the best connections in the biotech industry, the controversial entrepreneur has pieced together a $40 million financing round for his new startup Flex Pharma. And he did it by tapping a long lineup of investors, including some high profile individuals in the Boston area.

Westphal assembled his brain trust for Boston-based Flex earlier in the year, which is pursuing new treatments for muscle cramps with the help of Harvard's Bruce Bean. Nobel Laureate Rod MacKinnon, a scientific co-founder, has helped supervise a study in healthy volunteers for its lead drug, which is aimed at stopping severe cramping caused by hyperactive nerves. Efficacy studies are now slated to get underway next year, with the financial support of former Apple CEO John Sculley, Alnylam CEO John Maraganore and a big cast of supporters.

Westphal's own Longwood Fund joined in with Bessemer Venture Partners, EcoR1 and Alexandria Equities, which developers facilities for biopharma companies, on the round. And the VIP list also included Bain Capital Managing Director (and Celtics co-owner) Stephen Pagliuca and Harvard professor Robert Kaplan.

Westphal had been CEO of Verastem ($VSTM) up until about a year ago, when he handed the reins over to president Robert Forrester. He is perhaps best known for launching Sirtris, which GlaxoSmithKline ($GSK) bought out for a whopping $720 million. The pharma giant then shut down the unit in 2013 after making little headway in the clinic, after generating a raucous controversy over the science involved. GSK then absorbed the R&D work into a larger research unit in Pennsylvania.

Westphal went on to run GSK's SR One venture group for a short stint and started up the Longwood Fund.

45

FierceBiotech
THE BIOTECH INDUSTRY'S DAILY MONITOR

Westphal's Verastem slashes staff in the wake of a clinical trial flop

October 9, 2015 | By John Carroll

Two weeks after Verastem was forced to acknowledge that its lead cancer drug had flopped badly in a clinical study for mesothelioma, founder and Executive Chairman Christoph Westphal has brought out the ax and chopped half the staff. Verastem ($VSTM) acknowledged the move after the markets closed on Thursday, hours after word of the cuts began buzzing throughout the Boston biotech hub.

In a statement, the company confirmed the word on the street, announcing that 20 staffers had been terminated. News of the trial failure in late September had put the biotech's shares into meltdown mode, losing more than 60% of its value in minutes.

Termeer and Westphal out as Verastem looks to end losing streak

by *Ben Adams* | Jun 15, 2016 7:19am

The struggling biotech Verastem ($VSTM) whose stock has nose-dived over the past 12 months has seen its executive chairman Christoph Westphal and director Henri Termeer--two major players in the industry--resign without pre-warning.

Flex Pharma falls on cramp treatment update

by *Ben Adams* | Oct 13, 2016 8:25am

Flex Pharma ($FLKS) saw its shares tumble by 17% premarket this morning on the news that its med had hit trouble in its latest study, although it did have some better news from its chat with the FDA.

Last year the biotech, founded by serial biotech entrepreneur Christoph Westphal, pulled off a $86.4 million IPO--above its expected range. Investors were not happy first thing this morning however, with its shares falling premarket by 16.9% to $9.30 a share.

46